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A Framework for Managing Agentic Lifecycle Assets inspired on ML Model Management

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Executive Summary

As Large Language Model (LLM) deployments at scale move beyond experimental testing environments and into fully autonomous "Agentic" systems, organizations are faced with an operational paradox that is fundamentally different than traditional systems management. Organizations are unable to create sufficient structures for managing, securing or trusting LLMs that can be deployed on a commercial basis. The paradigms traditionally used to manage models (e.g., MLOps) were developed based on the premise that models are static ("stateless") and therefore cannot support the dynamic, multi-stage reasoning processes required by autonomous agents. Therefore, this study addresses the strategic gap by providing a new validated framework that allows organizations to systematically convert Agentic AI from a "black box" process to a useful enterprise asset.

The construction of this framework is based on a two-stage mixed-methods research design. The first stage, consisting of 32 enterprise experts doing a Choice-Based-Conjoint survey (CBC), used the resulting utility as input to generate a data-driven framework, which reflected real-world needs of AI practitioners rather than being based upon speculative theory. The purpose of the second stage was to identify the major impediments to the widespread adoption of AI from the perspective of an organization's senior AI practitioners, via a set of qualitative “think-aloud” interviews.

The empirical findings from this study have demonstrated an extreme change in the professional standards for AI maturity. The most dramatic finding was that Time to Value (TTV) is top indicator of success. The utility value for TTV at +26.54 was significantly larger than technical accuracy, with participants viewing technical accuracy as a baseline commodity. Further, the research found a "Governance Paradox", as professionals were overwhelmingly

negative towards implementing HITL (human-in-the-loop), citing it as one of the major causes of "unobserved procedural errors." Conversely, the data indicate a requirement for "logic-as-code," where the user prompt and schema of tools are treated as first class versionable artifacts that are managed with the same level of discipline as their binary versions of software.

These tensions can be resolved using the Validated Agentic AI Management Framework, which uses the concept of "Guardrailed Autonomy" introduced in this paper. This concept states that enterprise safety should not rely on human bottlenecks, but instead surround autonomous agents with unbreakable technical execution boundaries. Through a combination of a Fragmented Asset Registry for tracking evolving agent logic and an Automated CI/CD Pipeline for validating behavior within a Context-Aware Least-Privilege security guardrail, the framework enables agents to pursue complex objectives at maximum speed.

The strategic implications of this work are in establishing some "Rules of the Road" applicable in the next generation of human-computer interactions. In using the Guardrailed Autonomy concept, executives have the ability to address the "Software Gap" and eliminate the barriers to observing behavior. Additionally, as an enterprise builds trust on the basis of auditable financial and resource utilization, it has a foundation upon which to develop its autonomous systems and grow them at a reasonable rate. Ultimately, this work is providing the operational infrastructure useful to realize the significant potential of Agentic AI by transforming the theoretical possibilities of such AI into actualized business performance.

Abstract

The enterprise migration toward Agentic AI is creating an "operability paradox." While enterprises are increasingly requiring their systems to autonomously scale, there currently exist no standard management frameworks for managing those systems. It is necessary to develop and implement common governance for these systems in order to ensure security, financial control, and the ability to move AI projects from pilot status to full-scale business operations. Therefore, this dissertation addresses this gap by developing and testing a new management framework aimed at achieving "Guardrailed Autonomy" in an enterprise setting. Qualitative "think-aloud" interviews were conducted with senior practitioners to obtain personal opinions and quotes to ground the quantitative data. By completing a Choice-Based Conjoint (CBC) study consisting of 32 enterprise AI professionals, this paper is able to quantify the relative importance of specific factors related to practitioner's trust in deploying autonomous AI solutions. The results indicate that practitioners place a strong emphasis on achieving "Time-To-Value" (+26.54 part-worth utility) above other considerations such as technical accuracy and have an aggressive aversion to the need for humans to intervene ("Human-In-The-Loop", -12.88) in favor of automated Context-Aware Security Guardrails (+19.55). The results of this dissertation provide evidence that enterprise AI maturity can be achieved by exchanging procedural human-gates for unbreakable technical-execution boundaries which enable maximum velocity. Overall, the research provides a validated roadmap for transitioning Agentic AI from a potentially risk-prone R&D experiment to a scalable, reliable and high-speed business resource.

Keywords: Agentic AI, AgentOps, AI Lifecycle Management, Choice-Based Conjoint Analysis, Time-to-Value (TTV), Guardrailed Autonomy

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1. Introduction

The evolution of artificial intelligence has evolved from passive or simple "chatbots" to "Agentic AI", which is defined as systems that can independently reason and execute multi-step business processes (Joshi, 2025). With this transformation comes a "Operability Paradox," where companies would love to have the ability to use autonomous agents in their operations; however they are unable to do so due to a lack of operational and security frameworks to allow them to properly utilize these types of systems. MLOps was developed to be used with static models, and so it does not account for the dynamic and state-full characteristics of agentic systems (Schlegel & Sattler, 2023). Due to the lack of a specific management framework for managing autonomous systems, these currently exist in many organizations as "black boxes", creating numerous risk factors relative to both organizational security and budgets, thereby keeping most high-potential AI projects in the experimental phase.

The purpose of this dissertation is to fill this gap with an empirically supported path toward “Guardrailed Autonomy,” which implies that the best method to control AI systems isn’t through slowing them down using human oversight approval mechanisms, but through implementing “break-proof” technical barriers to support maximal velocity while maintaining total auditability. In order to determine such an approach the research employs a two-phased methodology: 1) Quantitative Choice-Based Conjoint analysis (CBC) to establish mathematical representations of the trade-offs that users are willing to undertake when evaluating speed, security and autonomy; 2) Qualitative think-aloud interviews with senior industry professionals to identify particular operating pain-points. The overall research is directed by the following Research Question (RQ):

- How can existing MLOps principles be adapted to create a robust *Agentic AI Model Management Framework*, and what specific components are considered most critical by industry experts (as validated by Conjoint Analysis) for reducing the Time-to-Value and enhancing the governance of enterprise-grade AI agents?

To answer this, the study pursues three primary Research Objectives:

1. **Identify** the operational frictions preventing practitioners from deploying agents at scale.
2. **Quantify** the relative importance of management features, such as security boundaries and asset tracking, through trade-off analysis.
3. **Validate** a framework that replaces manual oversight with automated "guardrails."

This paper makes two types of contributions; academic and practical. This paper extends the accepted principles of DevOps to the autonomous age, and presents a validated Agentic AI Model Management framework that will allow leaders to transition from a potentially risk-prone R&D experiment to a scalable, reliable and high-speed business resource.

The remainder of this work explores the current theoretical landscape of autonomous governance and details the methodology used to capture practitioner insights. The thesis then presents the empirical results and concludes by synthesizing these findings into the final validated framework, along with a discussion on the future of agentic operations.

2. Literature Review: Agentic AI Model Management Framework

2.1 MLOps Foundations and the "Software Gap"

Modern development of Artificial Intelligence depends upon the use of Machine Learning Operations (MLOps) -- an area of study that was developed for the purposes of resolving both the "management crisis" and "analyst amnesia" which are intrinsic within

unstructured data science (Vartak et al., 2016). Although some earlier frameworks have established standards for separating model training from its deployment so as to allow for reproducibility (Chen et al., 2020), contemporary taxonomies have revealed a continued "Software Gap." This gap distinguishes highly mature "Model Management" (binaries) from fragmented "Software Artifact Management," where the handling of code, configurations, and environmental dependencies often drives practitioners toward inefficient, ad-hoc versioning strategies (Schlegel & Sattler, 2023; Idowu et al., 2024). Day 1 MLOps, operationalization of MLOps capabilities (i.e. automated training), is highly represented, however, platforms providing Day 2 capabilities such as monitoring model performance and detecting model drift are not, with less than 40 percent platforms offering these services (Berberi et al., 2025).

2.2 The Anatomy and Governance of Agentic AI

The shift from static models to autonomous, compound systems is defined by "Goal Pursuit" and "Environmental Interaction"(Brohi et al., 2025). The following table synthesizes the architectural pillars required for agentic autonomy:

Table 2.2 - Technical Characteristics of the Components that Define Agentic AI

System Component	**Technical Characteristics**	**Ref.**
Cognitive Core	Transitions from passive mapping (X→Y) to autonomous "Goal Pursuit" and environmental interaction.	Brohi et al. (2025)

Cross-Layer Architecture	Comprises the Profile (roles), Memory (vector stores), Planning (ReAct/CoT), and Action (tool execution).	Adabara et al. (2025)
Interoperability	Facilitates agent-to-agent (A2A) handshakes via "Agent Cards" and the Model Context Protocol (MCP).	Ehtesham et al. (2025)
Governance Boundary	Shifts from model-centric to "Capability-Centric" risk assessment based on permission boundaries.	Khoo et al. (2025)

The emergent nature of these systems introduces "autonomy related hazards" that requires a paradigmatic shift from deterministically defined pipeline to risk management approach centered around "Goal Drift" and "Permission Boundaries" (Allam & Dempere, 2025; Khoo et al., 2025).

2.3 The Operational Frontier: AgentOps and GitOps

AgentOps represents a new iteration in MLOps that focuses on cost per goal metrics as well as stateful session management (Joshi, 2025). A significant gap between evaluation metrics exist since most frameworks assesses a model's performance based on its ability to make predictions as opposed to how safely it will execute them (Adabara et al., 2025; Shukla, 2025).

The research indicates the need for adaptive monitoring and “session replay” technologies to provide real time traceability of complex decision making processes (Berberi et al., 2025; Shukla, 2025).

The "Provenance Gap", the inability to identify the specific prompt or tool version active during a failure, that exists must be closed (Schlegel & Sattler, 2023) by implementing and using GitOps, along with "Infrastructure as Code" (IaC) concepts to enable greater deployment speed and to establish "Shift-Left" Governance, where compliance occurs prior to production versus after, through use of automated compliance testing as part of the production process (Debroy et al., 2018; Latendresse et al., 2024).

2.4 Lean Governance and the Organizational Trust Barrier

AI implementation has to be executed on a technological level using the "Value Stream" paradigm that uses work flow, continuous feedback, and eliminating waste ("Muda") to achieve reliability. Reliability can be measured through use of DORA metrics such as Change Failure Rate, which will measure how reliable an organization's processes are when it comes to developing AI assisted projects (Joshsi, 2025; Forsgen, et.al. 2018). A standardized framework for governance can provide business acceleration capabilities by providing "swim lanes" that define specific liability requirements and allow organizations to implement "light touch" automated compliance requirements into their CI/CD pipelines (IMDA, 2026; Tashkinov, 2024).

The last impediment to adoption is "trust calibration," which involves helping users identify agent success boundaries through natural language Explainable AI (XAI). In order to preclude "moral crumple zones", where humans may be unfairly blamed for the errors made by autonomous systems, frameworks need to include explicit "kill switches" along with unambiguous end-user responsibility assignment (Adabara et al., 2025; IMDA, 2026). For there

to be successful adoption, there has to be a "human-technology-organization" (HTO) shift and view autonomous agents as co-pilots, and not replacements, while providing workers with a sculpture of physiological safety (Tashkinov, 2024).

3. Materials & Methods

3.1 Drafting an Initial Framework - Ingredients

The methodology of this study turns the theoretical gaps found in the literature into quantifiable variables through the use of five core attributes that have three mutually exclusive levels. The rationale behind this design was to force Subject Matter Experts (SMEs) to make a choice among less than ideal options; therefore, they would be forced to make critical decisions about how to balance competing values related to technological autonomy, governance rigor, and speed of deployment (Joshi, 2025). The five core Attributes and their different Levels are defined as follows:

1. **AI Asset Registry Scope**: Evaluates the transition from traditional MLOps (static model binaries and datasets) to "Unified Agent Identity" bundles encompassing iterative prompts, tool configurations, and active memory states (Chen et al., 2020; Schlegel & Sattler, 2023), with Fragmented Asset Registry as the midpoint.
2. **Governance and Deployment Approval**: Weighs operational friction, ranging from strict human-in-the-loop (HITL) manual sign-offs to dynamic rule-based gates, to finally fully automated CI/CD pipelines (Kim et al., 2013; IMDA, 2026).
3. **Post-Deployment Observability**: Shifts focus from static output metrics (accuracy/latency) to a next level of resource utilization, and as the third level

"Goal Drift" monitoring and step-by-step reasoning traces (Joshi, 2025; Shukla, 2025).

4. **Security and Execution Boundaries**: Addresses psychological barriers to adoption through technical controls, with isolated sandboxing as a first level, just-in-time (JIT) permissions as the second, to "Context-Aware Least-Privilege" autonomy as the third (Adabara et al., 2025; IMDA, 2026).
5. **Primary Business Performance Metric**: Weighs technical accuracy (low error rate) against operational velocity, measured by DevOps metrics such as "Lead Time for Changes", and finally the broader "Time-to-Value" (TTV) and end-user adoption metrics (Forsgren et al., 2018).

3.2 Conjoint Survey Design

A Choice-Based Conjoint (CBC) survey was chosen to reflect the cognitive burden associated with designing an enterprise IT solution, as opposed to having respondents simply rank the choices; the survey forced respondents to make real-world tradeoffs rather than simply ranking their preferences (Gell, 2023). The survey took place using the Sawtooth Software platform. The software used a Randomized Choice-Task Algorithm in order to randomly display 8-12 different screens of two competing framework profiles that were displayed side by side. This allowed for the calculation of both individual Part-Worth Utilities and Overall Attribute Importance.

3.3 Data Collection and Analysis Strategy

This thesis uses a hybrid, mixed-methods approach to compliment mathematical preferences with thought out rationals. Data collection took place in two phases:

- **Quantitative (Phase 1)**: Digital distribution of the CBC survey to a validated sample of industry professionals to achieve statistical significance.
- **Qualitative (Phase 2)**: Supervised "Think-Aloud" sessions with eight senior practitioners who verbalized their internal justifications while completing the survey, capturing the organizational context and risk factors driving their trade-offs

*Table 3.3 - **Data Synthesis Matrix**. At a glance view of the data source along with their analytical method and primary output.*

Data Source	**Analytical Method**	**Primary Output**
Quantitative Conjoint	Hierarchical Bayes	Part-Worth Utilities & Importance
Qualitative "Think-Aloud"	Thematic Coding	Contextual Justifications

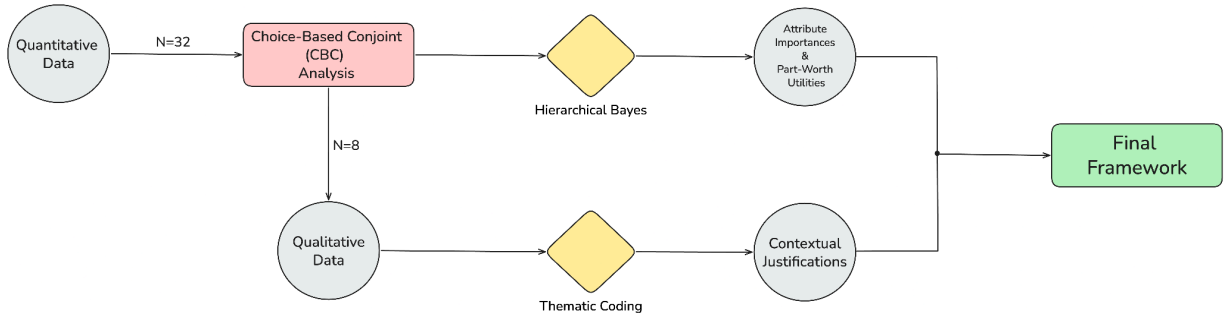


*Figure 3.3 - **Sequential Mixed-Methods Research Design for Framework Validation**. The procedural architecture illustrating the synthesis of quantitative trade-off data and qualitative practitioner insights.*

Figure 3.3 displays the visual flow of the process by which both quantitative and qualitative data were synthesized to produce one overall, comprehensive analysis. In addition, Table 3.3 synthesizes tabulary the primary analytic techniques that were employed for each source in addition to illustrating the primary outputs from those techniques.

The use of a Hierarchical Bayes algorithm was applied for the purpose of quantitatively analyzing the responses from the survey to generate estimates of attribute importances & individual part-worth utility scores. Subsequently, the qualitative notes underwent thematic coding to provide identification of recurring operational constraints or "bottlenecks" associated with these same five attributes.

4. Results

4.1 Sample Composition and Professional Context

The final, proposed framework is grounded in the inputs of **32** SMEs with a pretty significant skew toward senior operational experience. The sample represents a cross-functional yet technical group required to bridge the gap between AI development and enterprise IT governance.

Table 4.1 - Participant Professional Profiles

	Category	**Count**	**% of sample**
Primary Role	Data Scientist	7	21.88%
	ML/AI Engineer	3	9.38%
	IT Management	2	6.25%
	Software Engineer	4	12.5%
	Technology Consultant	5	15.63%

	Other Technical Role	11	34.38%
Years of Experience	0-2 Years	8	25%
	3-5 Years	6	18.75%
	5-10 Years	9	28.13%
	10+ Years	9	28.13%

As shown in Table 4.1, the survey population exhibits decent bias toward those with technical execution responsibility and senior operational history. The majority of respondents (50%) have a role in technical aspects of AI engineering and/or architecture; greater than 50% (56.25%) of respondents also have greater than five years of work experience within the industry. The respondent characteristics ensure that the tradeoffs captured by the framework are relevant to "Day 2" high stakes enterprise operations as opposed to new or experimental environments.

4.2 Quantitative Conjoint Analysis: Attribute Importance

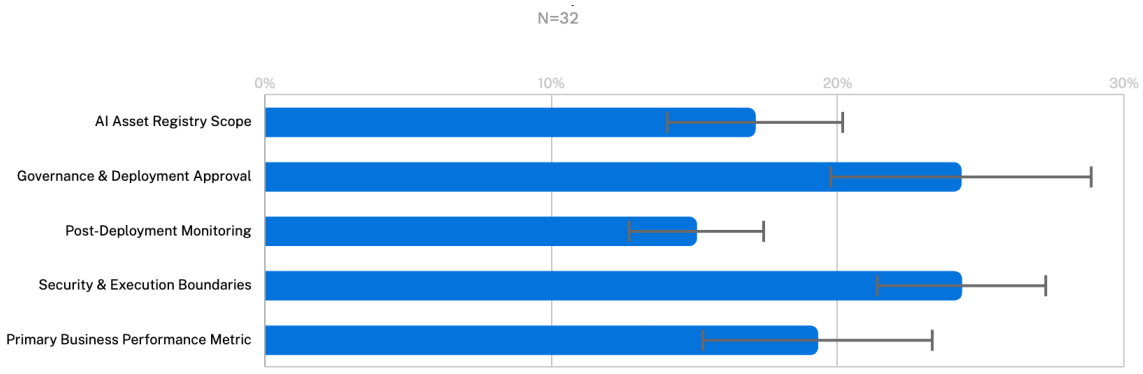


*Figure 4.2 - **Average Relative Importance of Agentic AI Framework Attributes**. A quantitative summary of the weighted preferences across the management dimensions derived from conjoint utility scores.*

To determine the high-level drivers of enterprise adoption, the conjoint data was aggregated to find the average relative importance of the five framework Attributes. As demonstrated by Figure 4.2, the analysis shows that there is a tie in statistical terms two main Attributes to drive technology adoption: Security and Execution Boundaries (24.30%), and Governance and Deployment Approval (24.28%): These two initial risk mitigation categories represent nearly fifty percent (48.58%) of all factors affecting practitioner's decisions. Third most important was the Primary Business Performance Metric (19.27%), followed by AI Asset Registry Scope (17.09%), while the aspects that would generally be considered “Day 2” techniques identified in literature of Post-Deployment Monitoring (15.04%) was the least important.

4.3 Granular Utility Analysis (Part-Worth Trade-offs)

While a hierarchical priority structure outlines overall preferences, a part-worth utility analysis provides an understanding of which operational configuration each expert prefers. Part-worth utilities are expressed as zero-centered values where positive values reflect preference for that particular attribute and negative values reflect operational "resistance" to changing that attribute. A full distribution of these mathematical trade-off analyses across the five identified framework attributes can be found in Table 4.3 (and a full visual breakdown in Appendix A). Each Framework Attribute shows its corresponding Attribute Importance, and the Utility Scores are highlighted in green if they represent the Tested Level with the highest score for that attribute, while the red highlights represent the lowest scoring Tested Level for that attribute.

Table 4.3 - Zero-Centered Part-Worth Utilities by Framework Attribute		
Framework Attribute	**Tested Level**	**Utility Score**
AI Asset Registry Scope (17.09%)	Baseline MLOps	-10.20
	Fragmented Tracking	**+13.28**
	Unified Agent Identity	-3.08
Governance & Deployment Approval (24.28%)	Strict Human-in-the-Loop	-12.88
	Dynamic Rule-Based Gates	+4.54
	Fully Automated CI/CD	**+8.34**
Post-Deployment Monitoring (15.04%)	Static Output Metrics	-16.55
	Resource Utilization	**+14.90**
	Goal Drift & Reasoning	+1.65
Security & Execution Boundaries (24.30%)	Static Sandboxing	+5.88
	Just-in-Time (JIT) Permissions	-25.43
	Context-Aware Least-Privilege	**+19.55**
Primary Business	Technical Accuracy	+0.34

Performance Metric (19.27%)	Operational Velocity	-26.88
	Time-to-Value (TTV)	**+26.54**

As shown by the data in Table 4.3, there has been a very strong reaction to manually implemented “Human-in-the-Loop” bottlenecks (-12.88) with the preference being for automated continuous integration/continuous delivery (CI/CD) processes (+8.34). In addition, as evidenced by the time to value utility score (+26.54), it appears that the ultimate metric of success is the fast resolution of business bottlenecks and not simply technical accuracy (+0.34) or the speed of loops. However, this automation is conditional on having Context-Aware Least-Privilege boundaries (+19.55), while the constant interruption of Just-in-Time (JIT) Permissions is the most heavily penalized security level (-25.43).

4.4 Qualitative Synthesis: Operational Barriers & Design Needs

The "Think-Aloud" sessions were analyzed and coded as part of contextualizing the utility scores so that the main drivers could be identified. A synthesis was performed to relate operational friction (mentioned by experts in the literature) to design requirements for an Agentic AI architecture. Codified detailed notes from this process can be found within the Appendix B.

Table 4.4 - Thematic Synthesis: Operational Friction vs. Design Needs

Attribute	**Primary Operational Friction**	**"Power Quote" Evidence**	**Design Requirement**
Governance	Human intervention	*“Human actions for*	Requirement:

	in deployment is viewed as a source of hidden procedural error.	*deployment usually lead to errors that aren't observable."*	Automated Guardrails & CI/CD Validation.
Business Metric	High technical accuracy is insufficient if the deployment process hinders speed.	*"Accelerate time to value without assuming too much risk."*	Requirement: Prioritize Time-to-Value (TTV).
Security	Security frameworks fail if they cannot detect internal logic shifts in real-time.	*"Security and even business KPIs fail if we cannot detect misalignment early."*	Requirement: Context-Aware Least-Privilege.
Asset Registry	Existing tools ignore the stateful, evolving nature of agentic components.	*"Agentic AI introduces fundamentally new objects: Prompts as first-class artifacts."*	Requirement: First-Class Versioning for Prompts/Memory.
Monitoring	Static benchmarks fail to capture the multi-step nature of	*"I prioritize observability of reasoning and goal*	Requirement: Reasoning Traces & Goal Drift Auditing.

	agent reasoning.	*drift over static accuracy."*	

The qualitative data in Table 4.4 reveals a fundamental shift in how practitioners perceive the risks of autonomous systems compared to traditional software. This synthesis highlights three critical operational transitions:

1. **From Manual Gatekeeping to Systematic Validation**: There is an glaring "Credibility Gap" in regards to human oversight. Experts specifically point out the manual "Human-in-the-Loop" approvals that are used by traditional governance models as points of failures in which "errors hide" because of operator fatigue and/or oversight. Therefore, the framework should be designed around Automated CI/CD Validations with the burden of safety being moved (shifted left) from the operator (operations) to the pipeline architecture itself (development).
2. **The Elevation of Non-Traditional Artifacts**: A key area of friction identified is the "second class" nature of prompts and memory with respect to how they are handled within current MLOps registry systems. From the practitioner's perspective, since agents are logic-based through the use of natural language, a given prompt should be treated as if it were executable code. To address this limitation, the proposed framework mandates that all prompts and memory snapshots will be treated as "versioned first class artifacts", providing users with the same ability to roll back and audit these objects as would exist for a user defined software object (e.g., a source file).
3. **Pragmatic Performance over Statistical Perfection**: The information illustrates why technical correctness was punished in the Quantitative Phase. Experts see agents through

a utility per unit of time prism. Therefore, if a system has a 99% success rate but takes months to deploy because of strict security protocols; then it will be seen as an operational failure. That said, the paradigm's shift toward Time-to-Value (TTV) as the lead KPI is justified if it can be supported by Context Aware Least Privilege Guardrails to catch logic misalignment in real-time.

In summary, the experts are not asking for less control, but for a different type of control, one that replaces the slow, manual "black box" monitoring of the past with a high-velocity, version-controlled execution model that allows for rapid debugging of reasoning traces.

5. Discussion

The primary challenge found in this investigation is the relationship between operational speed as required by an enterprise, and unpredictability as created by autonomous agents. The solution proposed in this research is the concept of "Guardrailed Autonomy." It represents a method of managing autonomy, that will allow for autonomous execution at high speeds while being technologically constrained to remain within an audited/defined boundary.

The culmination of this research is the Validated Agentic AI Management Framework (Figure 5.1), a structural blueprint designed to resolve the tension between autonomous velocity and enterprise liability. This model represents the "Guardrailed Automation" that moves beyond static, linear MLOps towards a continuous, high-speed execution loop.

5.1 Defining the Validated Management Framework

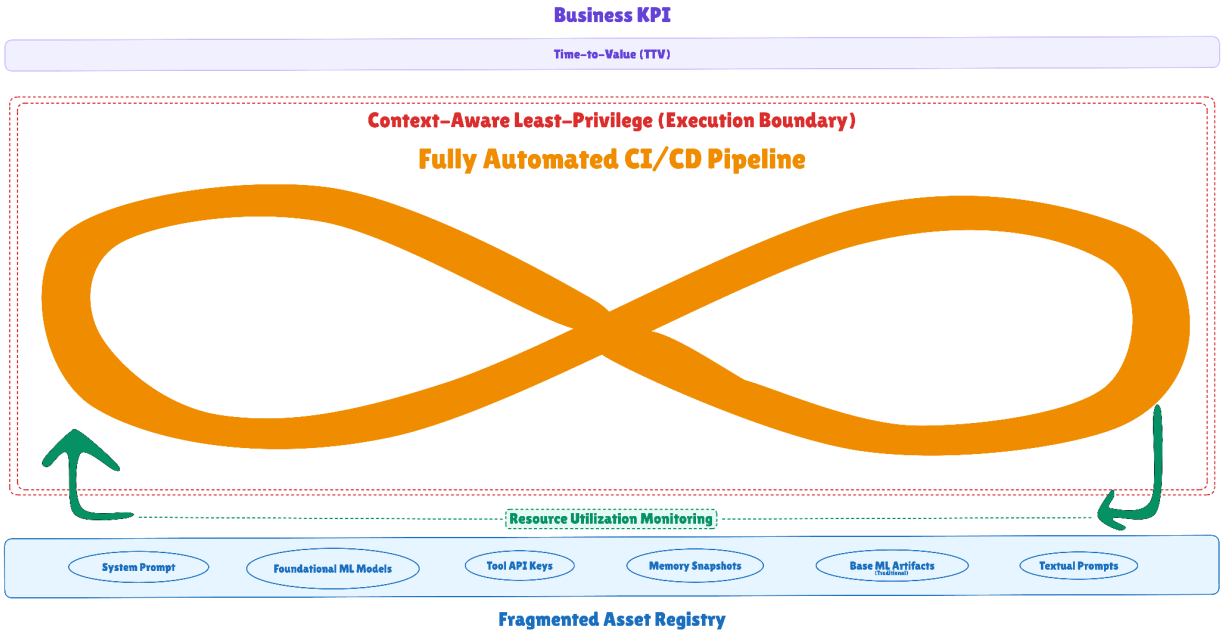


*Figure 5.1 - **Validated Agentic AI Management Framework**. The resulting managerial-level diagram from the attributes with highest importance*

5.2 Closing the Software Gap through Logic-Based Artifacts

The Blue Fragmented Asset Registry located at the base of the total structure represents the goal of controlling unorthodox assets with precision. By pulling the model, API tool key, and prompt up through the green arrow on the left to execute them within the main loop of the system, these artifacts become viewed by the framework as "First Class Artifacts." This structure directly remedies what Schlegel & Sattler (2023), called the "Software Gap"; it provides equal treatment to an agents logic (prompt) like what typical software code does with respect to versioning. A portion of the qualitative findings indicate that modularization was necessary because "Agentic AI introduces fundamentally new objects" that current MLOps tools do not track sufficiently well.

5.3 Resolving the Autonomy Paradox via Automated Integrity

The core of “Guardrailed Autonomy” is represented by the orange infinity loop (Automated CI/CD), encased by the red dotted line of the execution boundary. Practitioners chose this method because they felt “[...] human actions for deployment usually lead to errors that aren't observable”, therefore the framework replaces human sign-offs with technical guardrails. By using the first way of DevOps (Systems thinking), the red validation boundary will check if the agents’ context and permission are correct automatically before allowing them to run (Kim et al., 2013). Therefore, it’s not the developer’s “OK”, but rather an unbreakable “logic shell” where the agent can run at full speed and velocity without being slowed down by procedure.

5.4 Prioritizing Time-to-Value over Traditional Technical Accuracy

The purple Time-to-value (TTV) box at the top of Figure X represents the framework’s overall strategic mandate. For enterprise experts, Technical Accuracy is a “Baseline Commodity”, i.e. an entrance requirement rather than a differentiation factor. In this manner, the Framework adheres to DORA Metrics in prioritizing the equivalent of Lead Time for Changes and resolving bottlenecks quickly over statistical accuracy alone (Forsgren et al., 2018), validating the maturity of Agentic AI. Success is no longer determined by science projects based on accuracy but by business tools based on their capacity to "accelerate time to value without assuming too much risk".

5.5 Solving the Economic Barriers to Behavioral Observability

The down arrow from the “Monitoring” layer represents a sort of economic safety net. While thinking out loud and processing their thoughts, experts have said that they would prefer to focus on “goal drift”, and “reasoning traces”, but the framework dictates that the monitoring

of resource utilization will always precede the monitoring of execution in order to provide observable evidence of financial behavior to remove “muda” or waste i.e. runaway API cost (Kim et al. 2013), before achieving “trust calibration” which is required for wide spread enterprise acceptance (Cetinkaya & Kramer 2025). Professionals basically have an "ideal" goal they would like to optimize, but their day to day prioritizes something else.

5.6 Research Limitations

The study identifies a major disconnect between theoretical preferences and real-world economics. Resource utilization is quantitatively preferred by the practitioners today, but this suggests that present day infrastructure costs for LLM are currently obscuring future needs for monitoring "Goal Drift" behavior in autonomous systems. Therefore, the framework focuses on the quantitative preferred choice, which may become outdated by the time token costs decrease.

The sample does demonstrate a high level of technical experience and is representative of a unique sub-group of senior early adopters. However, because of this, the results indicating that some respondents aggressively rejected all forms of manual "Human-in-the-Loop" oversight may not be generalizable to those with lower levels of technology maturity in either private or public sectors who are subject to legal mandates requiring manual sign-off as opposed to these being organizational choices.

A drawback of Choice Based Conjoint (CBC), is that it relies solely on expressed preferences, rather than actual behavior. Think-aloud sessions can provide contextual information; however, respondents may demonstrate Social Desirability Bias, stating they support a certain idea in a research setting (such as the Guardrailed Autonomy concept) but return to less autonomous and less systematic ad hoc control when using this same product for a critical high-stake crisis.

5.7 Avenues for Future Research

Future research will have to define "Cost-per-Goal" as the principal AI version of DORA-style metrics of operational performance.

Practitioners established financial transparency as the base level of trust, there appears to be a significant need for enterprise management frameworks to close the technology execution vs. business Return On Investment (ROI) gap.

There seems to be a clear design need for "Lightweight Reasoning Traces." Research needs to explore how organizations may document an agents' alignment and reasoning logic using lower-cost computing power than current full-session replay solutions provide, in order to resolve the "conflict" between behaviorally safe practices and resource efficiencies.

6. Conclusion

This thesis was motivated by the "Operability Paradox," which describes how companies desire the capability of AI agents, yet have no viable means of managing those capabilities. This research provides a valid pathway for companies to transition from the experimental phase, to large scale implementation

At its core, this thesis defines and validates the concept of Guardrailled Autonomy, an autonomous capability that is both logically restricted and technically managed. Thus, for AI agents to become productive in a professional environment, safety mechanisms must exist and operate autonomously rather than being a time consuming and labor intensive manual processes. By automating boundary conditions, organizations may establish a logic shell around their agent(s), thereby ensuring they are operating within acceptable parameters.

The study also identifies two key paradigm changes in how we approach AI. Firstly this study shows that input for an artificial intelligence system is more than just a prompt or memory snapshot - it has become first class source code and should therefore be treated similar to any other application development file. Secondly the study shows how Time to Value has replaced accuracy of technology as the most important thing for companies today. In today's fast paced business environment the ability to quickly provide solutions to problems they face is what successful businesses are defined by.

Ultimately, this framework suggests "rules of the road," so to speak, for how humans are all going to work together with computers. Using automated guardrails allows companies' leadership to transition from seeing AI as experimental risk, to using AI as a viable and reliable asset in which they have confidence when making business decisions. It creates a foundational structure that sllows companies to develop trust that they can scale autonomous systems, and thus transform the potential of autonomy into actual results.

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Appendix

Appendix A - Full Visual Breakdown of the Individual Part-Worth Utility Scores

Table A.1 - ***Comprehensive Part-Worth Utility Estimates for All Attribute Levels****. A detailed visual breakdown of the individual preference scores assigned to every level within the five studied management dimensions.*

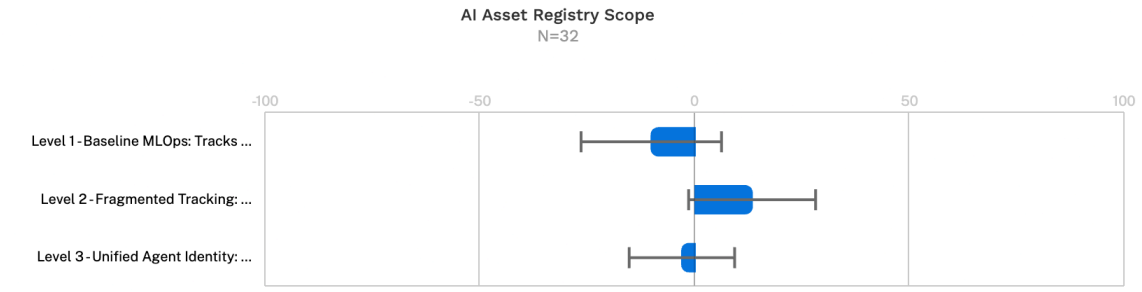


Figure A.1 - ***Part-Worth Utility Estimates for AI Asset Registry Attribute Levels****. Fragmented Tracking had the highest score, with Baseline MLOps (the traditional way) having the lowest.*

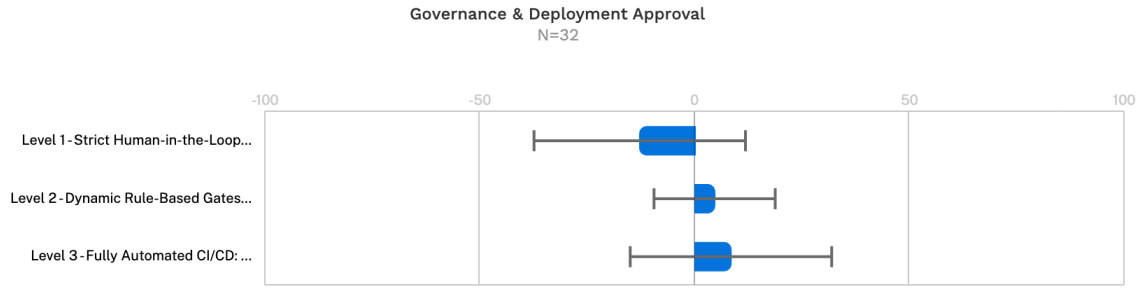


Figure A.2 - ***Part-Worth Utility Estimates for Governance & Deployment Approval Levels****. Fully Automated CI/CD had the highest score, with Strict Human-in-the-Loop having the lowest.*

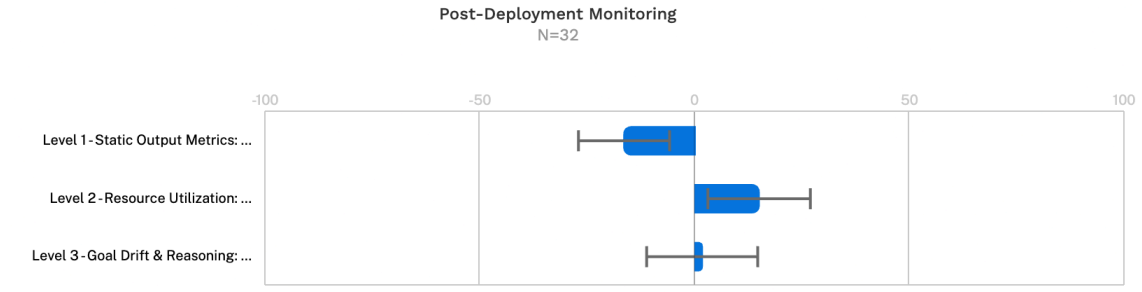


Figure A.3 - ***Part-Worth Utility Estimates for Post-Deployment Monitoring Levels****. Resource Utilization had the highest score, with Static Output Metrics having the lowest.*

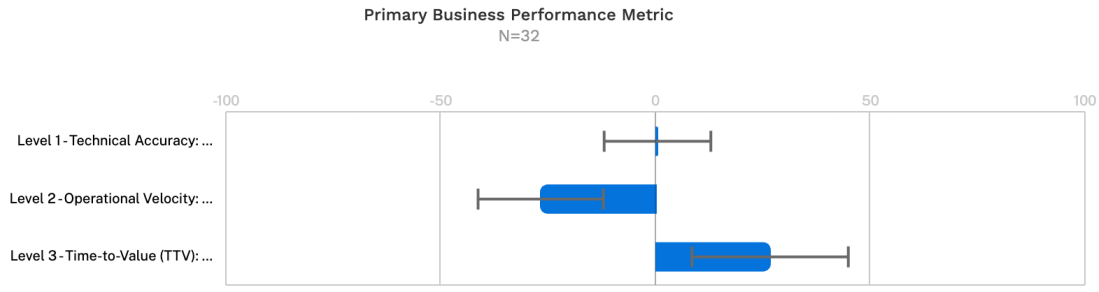


*Figure A.4 - **Part-Worth Utility Estimates for Primary Business Performance Metric Levels**. Time-to-Value (TTV) had the highest score, with Operational Velocity having the lowest.*

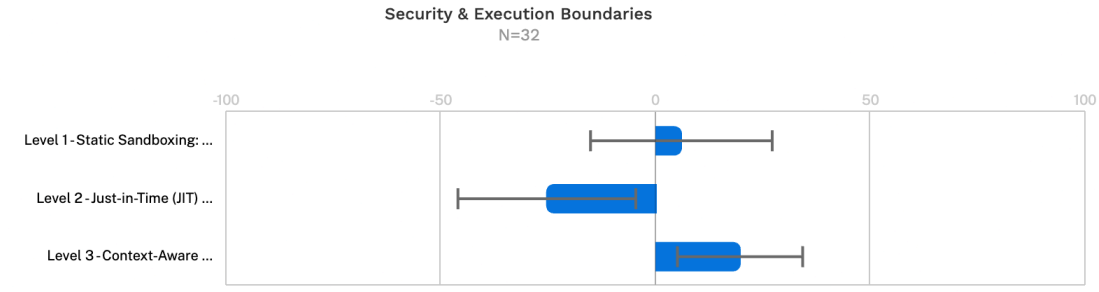


*Figure A.5 - **Part-Worth Utility Estimates for Security & Execution Boundary Levels**. Context-Aware Least-Privilege had the highest score, with Just-in-Time (JIT) having the lowest.*

Appendix B - Thematic Analysis of Qualitative Data

Table B.1 - Thematic Analysis of Qualitative Data - Tools and Priorities

Raw Participant Quote (Excerpted)	**Assigned Code / Theme**	**Corresponding Framework Attribute**
"HITL [Human-in-the-loop], Technical Accuracy are the top factors right now [...] Keeping human in the loop to supervise relevant moments of the AI agent"	Absolute Requirement for Human Supervision	Governance and Deployment Approval Workflows
"The single most important factor for me was the ability to understand, constrain, and correct agent behavior [...] Observability of reasoning and goal drift over static accuracy"	Need for Behavioral Tracking and Correction	Post-Deployment Observability and Monitoring
"Strong execution boundaries (least-privilege, approvals for risky actions) [...] Governance, security, and	Strict Access Control and Guardrails	Security and Execution Boundaries

even business KPIs eventually fail if we cannot detect misalignment early"		
"I prioritize control over deployment (meaning human oversight) [...] Having a human or at least strong approval gates keeps things safe while the tools are sill maturing."	Manual Approval Gates for Safety	Governance and Deployment Approval Workflows
"Most important factor is the security of data and validate outputs by humans, until the technology is fully functional."	Data Privacy and Output Validation	Security and Execution Boundaries / Governance
"Accelerate time to value without assuming too much risk [...] well balanced options that would cover 80% of our needs"	Pragmatic Efficiency	Primary Business Performance Metric
"Having human actions for	Automated but Observable	Governance and Deployment

deployment usually leads to errors that aren't observable [...] Making sure your CI/CD is properly configured, has tests, breakpoints, proper validations"	Pipelines	Approval Workflows

Table B.2 - Thematic Analysis of Qualitative Data - Tooling and Governance Insights

Raw Participant Quote (Excerpted)	**Assigned Code / Theme**	**Corresponding Framework Attribute**
"MLFlow works well for models and experiments, while agentic AI needs tracking of prompts, tools, memory [...] New platforms designed for agents will be needed."	Requirement for Dynamic Asset Tracking	AI Asset Registry Scope
"Traditional MLOps tools like MLflow are built around Static models [...] Agentic AI introduces fundamentally new	Shift from Static to Stateful Artifacts	AI Asset Registry Scope

objects: Prompts as first-class artifacts, Memory (stateful, evolving), Agent graphs / planners"		
"Because agents reason, plan, and take multiple actions, they are often 'black boxes.' MLflow Tracing provides deep visibility into the agent's execution flow [...] debugging multi-step reasoning"	Necessity of Reasoning Visibility	Post-Deployment Observability and Monitoring
"Stuff like MLflow still makes sense for the foundation [...] But once you move into agents, you're dealing with a system that can remember things, call tools, make decisions, and act... needs a new operating layer on top"	Foundational Extension for Tool Use	Governance / AI Asset Registry
"Tools like MLflow can be	Architectural Evolution for	Entire Management

adapted for parts of Agentic AI [...] but they don't fully support agent workflows like reasoning, memory, and tool use. So, we'll likely need new or heavily extended platforms."	Autonomy	Framework